

NGUYEN CUNG THANH

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EDUCATION

• Sophomore in VinUniversity

Bachelor of Science in Computer Science

- CPGA: 3.62/4.0, Dean's List (Fall 24, Spring 26).

September 2024 - Present

• Hanoi University of Science High School for Gifted Students

Top 1 high school for gifted student in natural science of Vietnam; Specialized in Mathematics

July 2021 - May 2024

PUBLICATIONS AND PREPRINTS

• Some Generalizations of a Geometric Problem from USA TSTST

Co-first Author; **Published** in The Mathematical Reflections; Supervisor: MSc. Tran Quang Hung - author of P2 IMO 2025 (HUS)

- Generalized the original problem, established complete proofs, and drafted the manuscript.

Jul 2024

[\[Paper\]](#)

• Logarithmic-Free Moment and Generalization Bounds for Uniformly Stable Algorithms

First author; **Preprint**; Supervisor: Asst. Prof. Nguyen Tuan Binh

- Prove the open problem in Uniformly Stable Algorithm.

August 2026

[\[arXiv\]](#)

• Inertia-Sensitive Kreiss Bounds for J -Selfadjoint Matrices

First author; **Under review** at Linear Algebra and Its Application (Q1); Supervisor: Asst. Prof. Nguyen Tuan Binh

- Construct and prove the whole paper including Hankel operator lemmas, the main theorem about Kress constant and find the family of matrix that prove the bound of the main theorem is sharp.

August 2026

[\[arXiv\]](#)

• How Finite-Rank Bottleneck Shape the Low-Rank Adaptation Landscape

Co-Author; **Under review** at NeurIPS 2026; Supervisor: Asst. Prof. Nguyen Tuan Binh

- Surveyed related work on low-rank adaptation and finite-rank bottlenecks, verified proofs and modified papers.

May 2026

[\[Research Page\]](#)

• Fast-Slow Reduction and Bifurcation Dynamics of a Spatial Intraguild Predation Model with Resource Refuge: Deterministic and Stochastic Robustness Analysis

Second Author; **Under review** at Computational and Applied Mathematics (Q2); Supervisor: Assoc. Prof. Nguyen Ngoc Doanh

- Analyzed equilibrium points and their stability in the proposed dynamical system and conducted theoretical analysis of the stochastic PDE model under Brownian perturbations.

May 2026

[\[Research Page\]](#)

- Implemented Maple simulations to validate analytical results and generate numerical figures and drafted paper.

• Asymptotic analysis, yield optimization and environmental noise in spatially coupled Gompertz populations

Co-Author; **Under review** at Applied Mathematics in Science and Engineering (Q1); Supervisor: Assoc. Prof. Nguyen Ngoc Doanh

- Analyzed equilibrium points and their stability in the proposed dynamical system.

May 2026

[\[Research Page\]](#)

- Implemented Maple simulations to validate theoretical predictions and generate numerical figures.

• On the Largest Prime Divisor of Polynomial and Related Problems

Independent Research; **Preprint**; Adviser: Assoc. Prof. Le Hai Khoi (VIASM)

- Developed new results on the largest prime divisors of polynomial values.

Mar 2025

[\[arXiv\]](#)

CONFERENCE PRESENTATIONS & SCHOLARLY ACTIVITIES

• Maximum Sustainable Yield in the Generalized Logistic Growth Model

First Author; **Presented** at International Statistical Conference (ISM VII), Malaysia

- Proved the main theorem about maximum sustainable yield in a generalized logistic growth model.

- Conducted Maple simulations to support theoretical findings.

Aug 2025

[\[Abstract pg.110, Slides\]](#)

• Problem 790, Pi Journal Math Challenge

Mathematical Problem Author; Topic: Analytic Number Theory

- Applied Mertens' theorem and analytic number theory techniques to create an original competition-style problem.

Mar 2024

[\[Problem\]](#)

ACHIEVEMENTS

• Silver medal in Algebra (Group A) at the Vietnam Mathematics Olympiad for Undergraduate Student

April 2026

• Silver medal in Calculus (Group A) at the Vietnam Mathematics Olympiad for Undergraduate Student

April 2025

• Consolation prize in the Vietnam National ICPC 2025

November 2025

• Participant of Vietnam Mathematics Olympiad

February 2023

RESEARCH EXPERIENCE

- **Research Assistant at Cosmos Lab** September 2024 - Present
Topic: mathematical modeling, dynamical system; Supervisor: Assoc. Prof. Nguyen Ngoc Doanh, Professor. [Pierre Auger](#) (IRD, Sorbonne Université)
 - Analyzed equilibrium structures, stability conditions of differential equation models.
 - Developed theoretical frameworks for maximum sustainable yield (MSY) using aggregation methods.
 - Proficient in using Maple for visualization and computation.
- **Research Assistant at BAIL** September 2025 - Present
Topic: Machine learning, Optimization, Probability Concentration Bound; Supervisor: Assist. Prof. Nguyen Tuan Binh (VinUniversity)
 - Weekly presentations on high-dimensional probability.
 - Applied techniques from linear algebra, optimization, and real analysis to construct machine learning theory, study training dynamics and matrix analysis.

PERSONAL PROJECTS

- **Personal Academic Website** [\[nguyencungthanh.github.io\]](#)
 - Built and maintain a personal academic website using Hugo, HTML/CSS/JS, deployed on GitHub Pages with automated CI/CD via GitHub Actions. Designed a modular publications system, mathematical blog with LaTeX rendering, full-text search, and responsive dark/light mode UI.
 - Features include bilingual support and tag/category taxonomy.

ACTIVITIES

- **Teaching Assistant** March 2024 - March 2025
Cmath Education (Specialized mathematics education center for secondary school students)
 - Teaching assistant of Teacher Vo Quoc Ba Can in the top-performing classes.
 - Tutored for three students who passed the specialized mathematics top high school entrance exam in Vietnam.
- **President of Vin Pioneers Club Gen 5** August 2025 - Present
The first and only math club at VinUni
 - Organized many meaningful activities, including Mentoring Session, Mocktest Exams, Integration Pi, selection VinUni Olympiad Math team and promoting Mathrathon
 - **Champion of Integration Pi** - A solving integral competition inspired by MIT and organized by Vin Pioneers

REFERENCES

- **Associate Professor [Nguyen Ngoc Doanh](#)**
 - Email: doanh.nn@vinuni.edu.vn
- **Assistant Professor [Nguyen Tuan Binh](#)**
 - Email: binh.nt2@vinuni.edu.vn